Derivatives: Power, Product, and Quotient Rules
Answer Key

AP Calculus AB/BC   Differentiation

Quick Answers

1. $20x^4 - 6x$
2. $-\frac{10}{x^3} + \frac{3}{\sqrt{x}}$
3. $3x^2$
4. $6x^2 - 10x + 6$
5. $\frac{13}{(x+4)^2}$
6. $-x^2 \sin(x) + 2x\cos(x)$
7. $-\frac{4x}{(x^2-1)^2}$
8. $3\sqrt{x} + \frac{1}{2\sqrt{x}}$, 6.25
9. $\frac{2x^2-6x}{(2x-3)^2}$, -4
10. $\frac{-x\sin(x)+(x+1)(x\cos(x)+\sin(x))}{(x+1)^2}$

Curriculum

Level: AP Calculus AB/BC   Differentiation
FUN-3.A–FUN-3.C – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 12 tagged checks.

1. $f(x) = 4x^5 - 3x^2 + 7$

Step 1. Differentiate term by term with the power rule $\frac{d}{dx}x^n = nx^{n-1}$: the exponent multiplies out front and drops by one.
Step 2. $4x^5 \to 20x^4$, $-3x^2 \to -6x$, and the constant 7 has slope zero.

$$\boxed{f'(x) = 20x^4 - 6x}$$

2. $g(x) = 6\sqrt{x} + \dfrac{5}{x^2}$

Step 1. The power rule needs powers, so rewrite: $g(x) = 6x^{1/2} + 5x^{-2}$.
Step 2. Differentiate: $6 \cdot \frac{1}{2}x^{-1/2} = 3x^{-1/2}$ and $5 \cdot (-2)x^{-3} = -10x^{-3}$.
Step 3. Rewrite with positive exponents.

$$\boxed{g'(x) = \frac{3}{\sqrt{x}} - \frac{10}{x^3}}$$

3. Limit definition for $f(x) = x^3$.
Step 1. Expand the binomial in the numerator: $(x+h)^3 = x^3 + 3x^2h + 3xh^2 + h^3$, so the numerator is $3x^2h + 3xh^2 + h^3$.
Step 2. Every surviving term contains h, so factor it out and cancel — this is the step the definition is really testing:

$$\frac{h\left(3x^2 + 3xh + h^2\right)}{h} = 3x^2 + 3xh + h^2.$$

Step 3. Now the limit is a substitution: let $h \to 0$ and the last two terms vanish.
Agreement. The power rule gives $\frac{d}{dx}x^3 = 3x^2$, the same result — which is exactly why the rule is allowed to replace this work.
$\boxed{f'(x) = 3x^2}$

4. $y = (x^2+3)(2x-5)$
Step 1. Product rule with $u = x^2+3$, $v = 2x-5$, so $u' = 2x$ and $v' = 2$.
Step 2. $y' = u'v + uv' = 2x(2x-5) + (x^2+3)(2)$.
Step 3. Expand and collect: $4x^2 - 10x + 2x^2 + 6$.
Check: expanding first gives $y = 2x^3 - 5x^2 + 6x - 15$, whose derivative is the same.
$\boxed{y' = 6x^2 - 10x + 6}$

5. $f(x) = \dfrac{3x-1}{x+4}$
Step 1. Quotient rule with $u = 3x-1$, $v = x+4$, so $u' = 3$ and $v' = 1$.
Step 2. $f'(x) = \dfrac{u'v - uv'}{v^2} = \dfrac{3(x+4) - (3x-1)}{(x+4)^2}$.
Step 3. Simplify the numerator: $3x + 12 - 3x + 1 = 13$. The x-terms cancel, which is why this derivative is never zero.
$\boxed{f'(x) = \dfrac{13}{(x+4)^2}}$

6. $h(x) = x^2 \cos x$
Step 1. Product rule with $u = x^2$, $v = \cos x$, so $u' = 2x$ and $v' = -\sin x$.
Step 2. $h'(x) = 2x\cos x + x^2(-\sin x)$. The minus sign comes from the derivative of cosine, not from the rule.
$\boxed{h'(x) = 2x\cos x - x^2 \sin x}$

7. $f(x) = \dfrac{x^2+1}{x^2-1}$
Step 1. Quotient rule with $u = x^2+1$, $v = x^2-1$, so $u' = v' = 2x$.
Step 2. $f'(x) = \dfrac{2x(x^2-1) - (x^2+1)(2x)}{(x^2-1)^2}$.
Step 3. The numerator is $2x^3 - 2x - 2x^3 - 2x$; the cubic terms cancel and $-2x - 2x$ remains.
$\boxed{f'(x) = \dfrac{-4x}{(x^2-1)^2}}$

8. $f(x) = (2x+1)\sqrt{x}$
(a) Step 1. Product rule with $u = 2x+1$ and $v = x^{1/2}$, so $u' = 2$ and $v' = \frac{1}{2}x^{-1/2}$.
Step 2. $f'(x) = 2\sqrt{x} + (2x+1) \cdot \dfrac{1}{2\sqrt{x}}$.
Step 3. Split the second term: $\dfrac{2x}{2\sqrt{x}} + \dfrac{1}{2\sqrt{x}} = \sqrt{x} + \dfrac{1}{2\sqrt{x}}$, so the two radical terms combine to $3\sqrt{x}$.
(b) Substitute $x = 4$: $3\sqrt{4} = 6$ and $\dfrac{1}{2\sqrt{4}} = \dfrac{1}{4}$, giving a tangent slope of 6.25.

$$\boxed{f'(x) = 3\sqrt{x} + \frac{1}{2\sqrt{x}}, \quad f'(4) = 6.25}$$

9. $f(x) = \dfrac{x^2}{2x-3}$
(a) Step 1. Quotient rule with $u = x^2$, $v = 2x-3$, so $u' = 2x$ and $v' = 2$.
Step 2. $f'(x) = \dfrac{2x(2x-3) - x^2(2)}{(2x-3)^2}$.
Step 3. Numerator: $4x^2 - 6x - 2x^2$, which collapses to $2x^2 - 6x$.
(b) At $x = 1$ the numerator is $2 - 6 = -4$ and the denominator is $(2-3)^2 = 1$.

$$\boxed{f'(x) = \frac{2x^2 - 6x}{(2x-3)^2}, \quad f'(1) = -4}$$

10. $f(x) = \dfrac{x \sin x}{x+1}$
Step 1. The outermost structure is a quotient, so start there: $u = x \sin x$ and $v = x+1$, with $v' = 1$.
Step 2. u is itself a product, so the product rule gives $u' = \sin x + x \cos x$.
Step 3. Substitute into $\dfrac{u'v - uv'}{v^2}$.

$$\boxed{f'(x) = \frac{(\sin x + x\cos x)(x+1) - x \sin x}{(x+1)^2}}$$